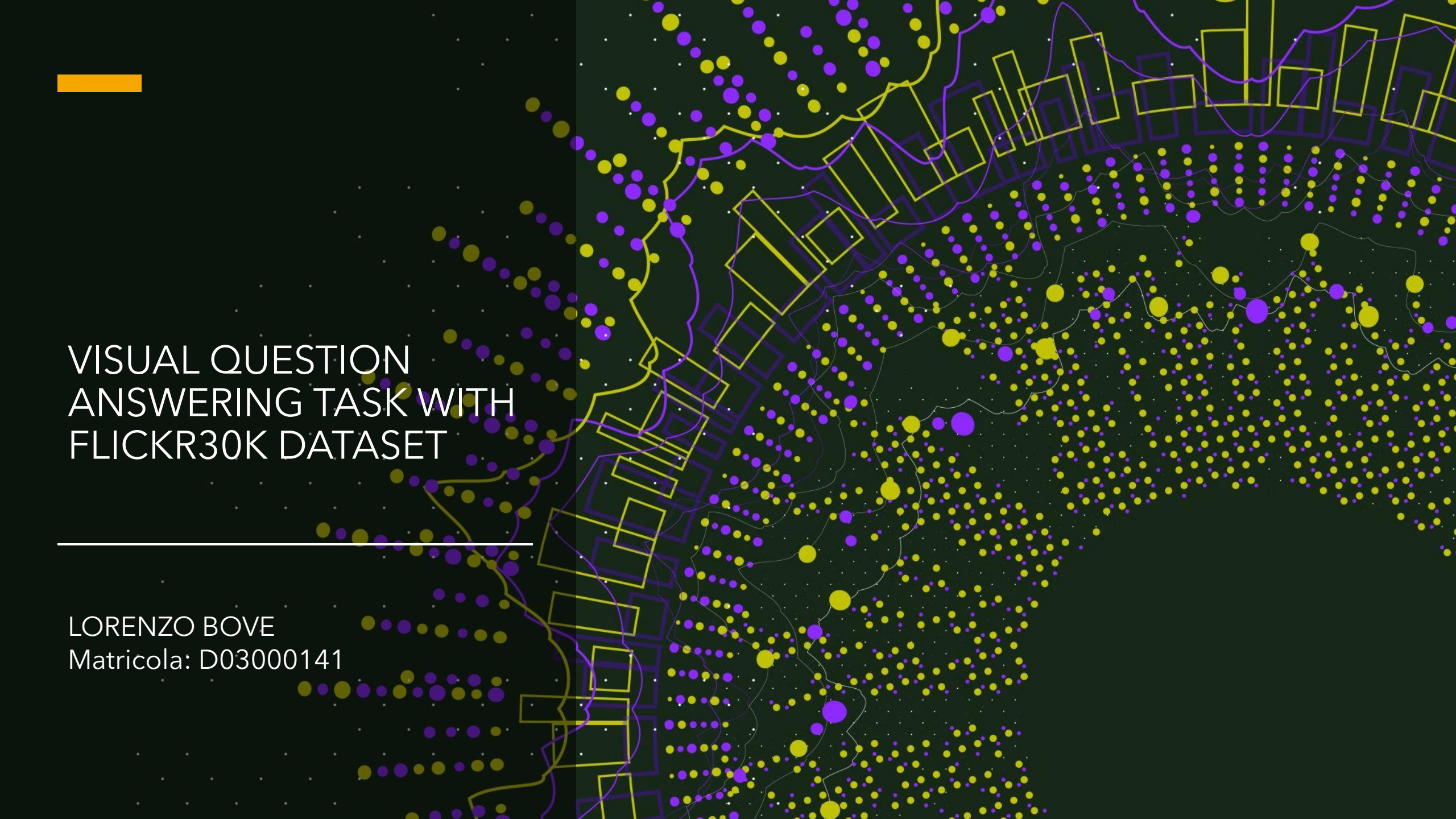




VISUAL QUESTION ANSWERING TASK WITH FLICKR30K DATASET

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Task Objective

- The goal of this task is to use the dataset **Flickr30k** to create a tool that is able to accomplish the **Visual Question Answering (VQA)** task.
- The Flickr30k dataset contains **31000 images** and **5 captions** for each image.

Setup and Methodologies

- **Setup**

1. GPU T4 X2 (KAGGLE)
2. LLAVA 1.5
3. 4000 images with 1 caption for image
4. DeepSEEK V4 Flash API

- **Methodologies:**

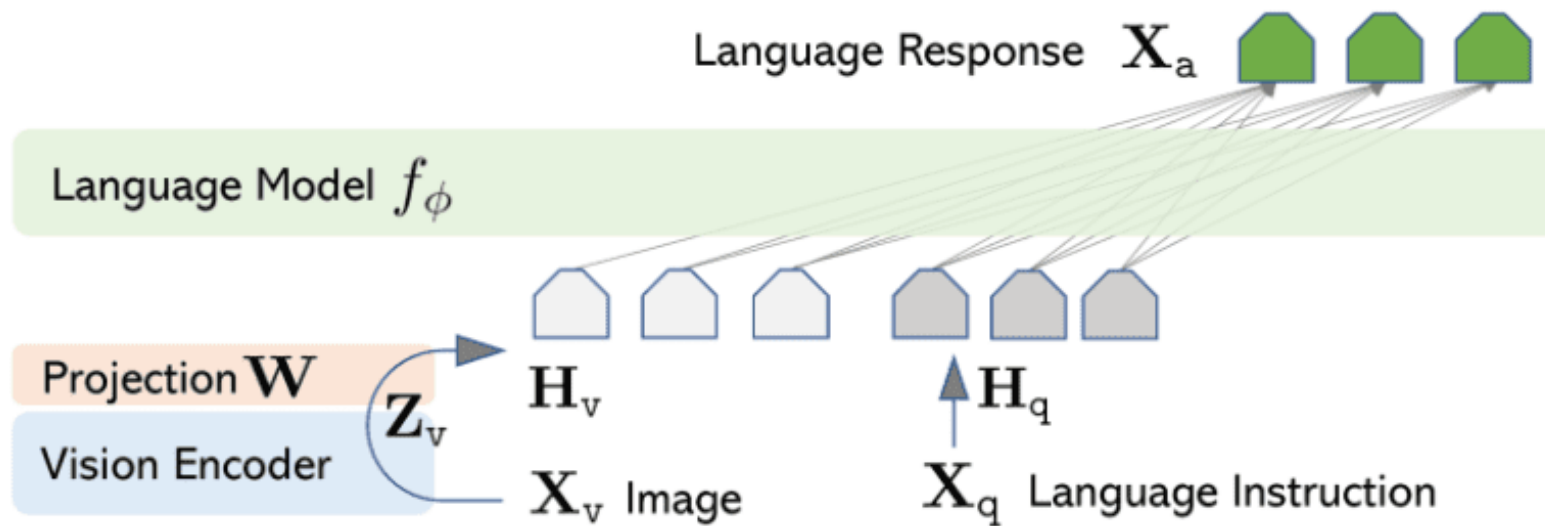
1. LoRA
2. BERTscore (evaluation metric)

Create a VQA Dataset

- The first step I did was to create a specialized VQA dataset from Flickr30K (not natively used for VQA)
- To do this I took the captions and give them as input to **DeepSeek V4** using the following prompt:
“Generate exactly 5 diverse question-answer pairs based ONLY on this caption. Cover different aspects: actions, locations, colors, objects, people, quantities. Do not hallucinate or invent details that are not present in the text. Answers must be short (max 10 words) and directly answerable from the caption.”

LLAVA Model

- The multimodal model that I used is LLAVA 1.5:





LoRA

- Since LLAVA is a pretrained model with 7 billions of parameters, and it is already able to do the VQA task, it is suitable to use LoRA to speed up the fine-tuning and to preserve the original knowledge of LLAVA.

1° Setup of parameters

- The first setup of parameters that I used to fine-tune the LLAVA model was:
 - N_epoch = 1
 - Learning rate = 0.0002
 - r = 16 (Rank of LoRA Matrix)
 - 4000 image/question pairs for train
 - 800 image/question pairs for test

Link to the model weights: <https://huggingface.co/Eikichi22/vqa-flickr30k-llava>

Results of 1° setup

- This set of parameters caused a problem regarding the answers that the final model gave as output.
- The training of the LoRA weights was too aggressive and this led the model to generate very random answer.
- The model learned to generate short answers without even looking at the image.

2° Setup of parameters

- To solve the previous problem, I tried to use lower values for those parameters:
 - N_epoch = 1
 - Learning rate = 0.00002
 - $r = 8$ (Rank of LoRA Matrix) <- in this way the matrix is smaller
 - 4000 image/question pairs for train
 - 800 image/question pairs for test
- Link to the model weights:
<https://huggingface.co/Eikichi22/vqa-flickr30k-llava-v2>

Results of 2° setup

- With a smaller LoRA Matrix and a lower Learning Rate the resulting model better achieved in the VQA task.
- Compared to the original LLAVA 1.5 model, the answer are shorter (thanks to the fine-tuning on Flickr30K dataset), but still the original model is more accurate at answering.



Evaluation

- To evaluate the resulting model I used the **BERTScore** metric.
- This metric is based on computing the distance between the embeddings of the real answer with the ones of the answer generated by our model.
- If most of the embeddings are near each other than the score will be high.

Conclusion

- The result of this homework is a model that has a lot of limitations about the VQA task. This happens because the model was fine-tuned only on a small part of the entire dataset and with only one epoch due to hardware limitations. However, the model is able to generate short answers thanks to the training on Flickr30k dataset.



LINK TO THE KAGGLE NOTEBOOK

<https://www.kaggle.com/code/lorenzobove24/progetto-speech-processing>